1 There are 12 boys and 15 girls in a class.
One boy and one girl will be selected to represent the class on the student council.
Work out the total number of ways of choosing a boy and a girl.

12×15

180

(Total for question 1 is 2 marks)

2 There are 17 boys and 14 girls in a choir.
One boy and one girl will be selected to sing a duet.
Work out the total number of ways of choosing a boy and a girl.

17×14

238

(Total for question 2 is 2 marks)

3 There are 14 boys and x girls in a choir.
One boy and one girl will be selected to sing a duet.
Taylor says there are 152 different ways of choosing a boy and a girl.

Could Taylor be correct?
You must show your working.

$14x = 152$

$x = 10.857\ldots$

Taylor cannot be correct. 152 is not divisible by 14.

(You cannot have 0.857 of a girl)

(Total for question 3 is 2 marks)

4 There are 5 starters and 6 main courses in a restaurant.

Work out the total number of ways of choosing a starter and a main course.

5×6

30

(Total for question 4 is 2 marks)

5 There are 4 starters, 7 main courses and 4 desserts in a restaurant.

Work out the total number of ways of choosing a starter, a main course and a dessert.

$4 \times 7 \times 4$

112

(Total for question 5 is 2 marks)

6 There are 5 starters, 6 main course and x desserts in a restaurant.

Riley says there are 130 different ways of a starter, a main course and a dessert.

Could Riley be correct?
You must show your working.

$30x = 130$

$x = 4.\dot{3}$

Riley cannot be correct. x would have to be a whole number.

(Total for question 6 is 2 marks)

7 A meal deal includes a sandwich and a drink.
There are 5 sandwiches and 7 drinks to choose from.

Work out the total number of ways of choosing a sandwich and a drink.

5×7

35

(Total for question 7 is 2 marks)

8 Mr Idris has 5 pairs of trousers, 9 shirts and 3 ties.

Work out the total number of ways of choosing a pair of trousers, a shirt and a tie.

$5 \times 9 \times 3$

135

(Total for question 8 is 2 marks)

9 There are 8 sandwiches and x drinks to choose from for lunch.

Pat says there are 96 different ways to choose a sandwich and a drink.

Could Pat be correct?
You must show your working.

$8x = 96$

$x = 12$

Pat could be correct. x could be 12.

(Total for question 9 is 2 marks)

10 There are 52 cards in a deck.
Peter is going to give one card to Casper and one card to Kelly.

How many different ways are there of going this?

52 x 51

2652

(Total for question 10 is 2 marks)

11 There are 52 cards in a deck.
Angel is going to give one card to Ben and one card to Chris and one card to Dylan.

How many different ways are there of going this?

52 x 51 x 50

132600

(Total for question 11 is 2 marks)

12 There are 52 cards in a deck.
Tom is going to give two cards to Jay.

How many different pairs of cards could Jay get?

$$\frac{52 \times 51}{2}$$

1326

(Total for question 12 is 2 marks)

13 There are 30 students in a class.
Two students are going to be selected to receive a prize.

How many different pairs of ~~two~~ students could be selected?

$$\frac{30 \times 29}{2}$$

435

(Total for question 13 is 2 marks)

14 There are 10 teams in a football league.
Two teams are going to be chosen at random to play a match.

Work out the number of different matches that could take place.

$$\frac{10 \times 9}{2}$$

45

(Total for question 14 is 2 marks)

15 There are 8 teams in a competition.
Each team will play every other team once.

Work out the total number of games played.

$$\frac{8 \times 7}{2}$$

28

(Total for question 15 is 2 marks)

16 There are 10 people in a room.
Each person shakes each other person's hand once.

Work out the number handshakes that take place.

$$\frac{10 \times 9}{2}$$

45

(Total for question 16 is 2 marks)

17 There are 20 people in a room.
Each person shakes each other person's hand once.

Work out the number handshakes that take place.

$$\frac{20 \times 19}{2}$$

190

(Total for question 17 is 2 marks)